Shrey Bansal

Senior Undergraduate, Department of Computer Science and Engineering

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Education

Georgia Institute of Technology Atlanta, US
Master of Science, Computer Science; GPA: 4.0 *2025 - Present*

Indian Institute of Technology, Kanpur Kanpur, India
Bachelor of Technology, Computer Science and Engineering; CPI: 8.9/10 *2021 - 2025*

Green Valley High School Vadodara, India
CBSE Class XII; Grade: 96.6% *2019 - 2021*

Navrachana School Sama Vadodara, India
CBSE Class X; Grade: 98.2% *2018 - 2019*

Scholastic Achievements

- Received the **Academic Excellence Award, IIT Kanpur** (Top 10%) for two consecutive academic years (2022, 23)
- Secured an **All India Rank 174** in **JEE Advanced 2021** among 142k students (2021)
- Secured an **All India Rank 238** in **JEE Main 2021** among 939k students (2021)
- **KVPY Fellowship**, securing **All India Rank 342(SX)** and **511(SA)** conducted by **IISc Banglore** (2020, 21)
- **State Topper, IOQP** - Among **Top 1%** in State - conducted by **IAPT** (2021)

Work Experience

Microsoft May'24 - Jul'24
Software Engineer Intern *Noida, India*

- Contributed to the PowerPoint Online Team by developing an **advanced slide editing canvas** for the WebApp
- Engineered a **feature-rich text editor** with functionalities like superscript, subscript, pattern fill, and strike-through
- Managed the **end-to-end text editing pipeline**, including user input detection, data processing, and rendering effects
- Collaborated with **international teams** to develop and deliver resilient code with **multiple fallback** mechanisms

Vibinex Jun'23 - Aug'23
Software Development Intern *Remote*

- Implemented a **CI/CD pipeline** using **GitHub Workflows** to automate **deployment process** of the extension
- Utilized GitHub Workflows to **trigger the pipeline** whenever a **pull request is merged** into the main branch
- Developed a feature **in Rust** to assess the **impact of pull request changes** on a codebase's **dependency graph**, utilising existing Language Servers to enable precise **evaluation of relevance** for a specific user

NOCIW May'23 - Aug'23
Full-Stack Android Development Intern *Remote*

- Engineered a **location-based** feature in a **React Native App** to dynamically identify and categorize nearby users with similar interests, enhancing user engagement by displaying personalized results
- Designed and upgraded various features to display local business' information on the **community platform**
- Made the feature for users to **claim rewards** based on their collected points, as well as view their upcoming rewards
- Developed a voice messaging feature for a chat application, storing audio as **base64 strings** on **AWS S3 Bucket**, and implemented real-time playback with waveform visualization to enhance user interaction

Publications

[1] S. Saklani, C. Goel, S. Bansal, Z. Wang, S. Dutta, T. M. Athawale, D. Pugmire, and C. R. Johnson. "Uncertainty-informed volume visualization using implicit neural representation". In *Proceedings of the IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024. doi: 10.48550/arXiv.2408.06018

Research Experience

Vajra Aug'25 - Present
Systems for AI Lab (SAIL), Prof. Alexey Tumanov *Project*

- Contributing to an in-house open-source **low-latency distributed LLM inference** serving system
- Built an **event-driven simulator** for **multi-tier prefix-cache** system; integrated **Vidur** as dynamic runtime predictor
- Added end-to-end support for **quantized models** like GPT-OSS in Vajra, utilizing **FlashInfer GEMM kernels**
- Working on Setu, a **Global Tensor Exchange system** for efficient tensor movement across cluster of GPU nodes

TokenSmith Aug'25 - Dec'25
Database Research Group, Prof. Joy Arulraj *Project*

- Worked on database improvements to efficiently query LLMs for **long-context Document Question-Answering**
- Leveraged database concepts like **indexing** and **graph-based** approaches to efficiently search answers from textbook
- Setup a **benchmarking framework** to evaluate implementation based on key metrics like **NLI Classification, semantic similarity, etc.**

Data Race Detection in GPUs Aug'24 - Dec'24
Undergraduate Research Project, Prof. Swarnendu Biswas

- Reviewed several papers to build a foundation for developing an approach to predict data races in GPU environments
- Utilized **NVBit** to generate program trace of micro-benchmarks and large GPU programs via binary instrumentation
- Designed and implemented an offline algorithm to detect predictable data races from the generated trace files
- Ensured soundness by testing on microbenchmarks and detected more races on larger benchmarks than SOTA detectors

Uncertainty-aware Volume Data Modeling and Visualization Jan'24 - Apr'24
Undergraduate Research Project, Prof. Soumya Dutta *Publication*

- Leveraged Uncertainty-Aware Deep Neural Networks to efficiently compress and store extensive volumetric data
- Incorporated **Residual Layers**, activation functions such as **SIREN**, and **LR Decay** in the network architecture
- Compared the accuracy and efficiency of three methods of compression: Deep Ensemble, MC dropout, and Evidential
- Utilized PSNR along with volume rendering and distance metrics within isocontour analysis to assess the accuracy
- Performed visual rendering of regenerated volumetric data using **ParaView** with uncertainty-based coloring

Teaching Experience

Graduate Teaching Assistant Aug'25 - Dec'25
CS 6422 (Database Systems Implementation, Georgia Tech)

- Assisted in the course organization by **managing assignment** submissions and grading
- **Cleared doubts** regarding course content and **led discussions** about the latest developments in database systems

Tutor Aug'24 - May'25
ESC 111/112 (Fundamentals of Computing, IIT Kanpur)

- Guided **60 undergraduates** in **C programming** through **tutorial sessions**, building practical problem-solving skills
- Assisted in **designing problem sets**, **coding labs**, graded **midterm/final** assessments to ensure fair evaluation

Key Projects

Agentic SRE System Jan'25 - Present
Self Project

- Architected a multi-agent system with master-slave topology using MCP servers for inter-agent communication
- Developed 16 specialized tools spanning intelligence, coding, research, and infrastructure categories
- Integrated Datadog, OpenTelemetry, SQL, and Kafka for log analysis, along with service dependency graph generation
- Implemented confidence-based consensus mechanisms across sub-agents for reliable multi-agent decision-making
- Enabled autonomous bug detection, remediation, and optimization in production-critical workflows

DSched - vLLM Oct'25 - Nov'25
Course Project (Systems for ML), Prof. Alexey Tumanov

- Integrated DSched, a compiler-driven ILP-based scheduler for DAG workloads, into vLLM as a custom prefill scheduler
- Modeled prefill jobs as STRL++ expressions over a resource-time plane for plan-ahead GPU placement
- Matched vLLM's heuristic scheduler on time-between-tokens and inter-token latency with modest TTFT overhead
- Instrumented per-request tracing via NVIDIA's NVTX library to capture queue wait time, execution duration, and preemption overhead

Procurement Intelligence Platform May'25 - Present
Self Project

- Built a production-ready vendor normalization pipeline using semantic embeddings and Milvus vector search with AWS Bedrock LLM validation for low-confidence matches
- Implemented cascading categorization to classify procurement transactions with fine-tuned ML bi-encoder for validation
- Developed Corvus, an AI procurement agent with 20+ tools for spend analysis, benchmarking, and vendor risk
- Developed Porter, a federal grant matching module with SAM.gov, automated vendor outreach, project management
- Designed the application in Next.js and React.js with AWS RDS PostgreSQL and pgvector for embedding storage

Pipelining MIPS Simulator Apr'25 - Apr'25
Course Project (Computer Architecture), Prof. Mainak Chaudhuri

- Developed a **pipelined 5-stage MIPS integer unit simulator**, from a multi-cycle, unpipelined design
- Implemented interlocked pipeline control to handle hazards, including stall mechanisms and phased execution
- Integrated a full bypass (forwarding) network within the simulation, reducing average CPI from **5.0 to 1.01**

Code Optimization Aug'24 - Nov'24
Course Project (Programming for Performance), Prof. Swarnendu Biswas

- Accelerated CPU performance with SIMD vectorization (**SSE4, AVX2**) and advanced loop transformations (permutation, tiling, unrolling, LICM), cutting runtime from **1.3s to 100ms**
- Optimized GPU kernels with CUDA and Thrust, achieving **16x speedup** for stencil and convolution patterns
- Built concurrent data structures (open-addressing hash table, lock-free stack) with fine-grained concurrency control

BtrFS MysticClones Feb'24 - Apr'24
Course Project (Linux Kernel Programming), Prof. Debadatta Mishra

- Implemented **Copy-on-Write (CoW)** for **page cache** of cloned files in Linux Kernel, focusing on **btrfs** filesystem
- Handled reading, writing, deleting, and truncating operations for cloned files, ensuring correctness and consistency
- Utilized inode locks to address **multithreaded scenarios** and **FIO** for benchmarking multithreaded operations
- Achieved notable **reductions in latencies** during file read and write operations, enhancing the overall performance

Sssnakelyzer Feb'24 - Apr'24
Course Project (Compiler Design), Prof. Swarnendu Biswas *Project*

- Developed a **Python to x86-64 compiler**, employing **Flex** and **Bison** as lexer and parser alongside **C++**
- Implemented support for static typing, string, lists, function calls and recursion along with OOPs and inheritance
- Translated 3AC Intermediate Code into **executable x86 Assembly** code, facilitating seamless execution with **gcc**
- Achieved top performance as a team by scoring **100% marks** across **all milestones** and **overall implementation**

GemOS Jul'23 - Nov'23
Course Project (Operating Systems), Prof. Debadatta Mishra *Project*

- Implemented system critical syscalls like **strace** and **ftrace** using **circular buffers** in GemOS
- Implemented memory management syscalls like **mmap, munmap** and **mprotect** as well as APIs for **malloc** and **free**
- Crafted lazy allocation strategies and developed vm area page fault handler in a 4-level paging system
- Implemented **cfork** syscall along with a **Copy-on-Write (CoW)** fault handler for efficient memory management

TripSay May'23
Self Project *Project*

- Developed a cutting-edge **travel expert AI**, utilizing **OpenAI API endpoints** within a **NextJS framework**
- Implemented a **user-friendly interface** that supports **text and audio input**, using the **Whisper API**
- Created a sophisticated chatbot capable of engaging in **natural language conversations**, allowing users to inquire about **various travel tips, generate personalized itineraries, and discover remarkable destinations**
- Utilized **Typescript** to enhance **reliability, ensuring type safety and improved tooling and integration**

E-Vaidya : HC Automation Mar'23 - Apr'23
Course Project (Software Development), Prof. Indranil Saha *Project*

- **Automated appointment procedures** at the Health Center at IITK with a **web-based application**
- **Separate interfaces** for doctors, nurses, patients, receptionists, and pharmacists designed using **ReactJS**
- Implemented features like **appointment scheduling**, medicine prescription, **pipeline to conduct appointments**
- Used **JWT Tokens** to provide authorization to each user and **bcrypt**, a NodeJS package to store hashed passwords
- Used **ExpressJS** for back-end, **NodeJS** as JS runtime environment, and **MongoDB** for Database Management

CSE - BUBBLE Mar'23 - Apr'23
Course Project (Computer Organization), Prof. Urbi Chatterjee *Project*

- Made a 32-bit **processor** CSE-BUBBLE with a **custom Instruction Set Architecture** using MIPS-32 as reference
- Used **Verilog HDL** to design and implement the ALU in a top-down approach based on the **custom ISA**
- Used the processor to write the MIPS code for **Bubble Sort Algorithm** and convert it into **Machine Code**
- Designed op-codes formats for efficient **instruction handling**, encompassing **ALU and branching** mechanisms

Space Travel System Mar'23
Course Project (Software Development), Prof. Indranil Saha *Project*

- Developed an **interactive command-line program** using C++ that implemented **OOPS concepts**
- Implemented a **role-based system** where users can choose among **3 roles**: Passenger, Astronaut, and Commander
- Designed a comprehensive system that allows passengers to **effortlessly book, view, and delete tickets**
- Employed appropriate **data encapsulation techniques** by declaring class members as **private or public**
- Promoted **code modularity and maintainability** by organizing the codebase into **separate files**

Neural Machine Translation Dec'22 - Feb'23
Brain and Cognitive Society (Winter Project), IIT Kanpur *Project*

- Developed a **Neural Machine Translation** model from scratch, for **English-Hindi language translation**
- Implemented an **LSTM Seq2Seq model**, with **encoder-decoder architecture** to capture **contextual info**
- Employed **tokenization techniques** to preprocess input sentences and **generated embeddings** using TensorFlow embeddings, enhancing the model's ability to **handle complex linguistic structures**
- Trained the model on **Kaggle dataset**, refining performance via **rigorous testing** to achieve **92% accuracy**

Echo: "ACCESS GRANTED!" Jun'22 - Aug'22
ACA, IIT Kanpur *Project*

- Utilized **Git and UNIX commands** to successfully overcome challenges in the OverTheWire-Bandit levels
- Employed **bash scripting** to **automate** routine tasks such as **data entry** and basic **statistical analysis**
- Explored **web components and networks**, to apply **web exploitation techniques** to **identify vulnerabilities**
- Utilized **FFuF**, a powerful tool for **brute-forcing**, to conduct comprehensive **web page searches**
- Explored **Assembly Language**, gaining insights into **program execution, registers, and function stacks**

Rover Base May'22 - Jul'22
SNT, IIT Kanpur

- Proficiently utilized **Autodesk Fusion 360** to create intricate **engineering and manufacturing designs**
- Developed a **Bluetooth-controlled car**, using **Arduino UNO and L298n Motor Driver** for seamless control
- Utilized **NVIDIA Jetson NANO** to build a rover with **depth sensing and autonomous braking systems**

MACHINE LEARNING : BEHIND THE CURTAIN Apr'22 - Jul'22
Stamatics, IIT Kanpur *Project*

- Acquired foundational knowledge of **Python, NumPy, Pandas library**, and concepts of **linear algebra**
- Developed a **linear regression model** using NumPy and Matplotlib to perform **binary classification tasks**
- Implemented **neural network with hidden layer** for classification using **NumPy, Pandas**, and **Matplotlib**

Positions of Responsibility

Technical Secretary Feb'23 - Sep'23
Association for Computing Activities, IIT Kanpur

- Organised events such as Happy Hour, Farewell, ACA Projects, and Summer School for CSE department students
- ACA Projects: Conducted intro sessions and projects allotment for IITK freshers and monitored project progress
- Adhyayan'23: Conducted a 4-week online Computer Science courses program in summers for PAN India students

Events Secretary Jun'22 - Apr'23
Entrepreneurship Cell, IIT Kanpur

- Organized the Annual Flagship event E-Summit'22 with a team of 20 secretaries.
- Hosted and managed the different competitions held by the Entrepreneurship Cell.
- Lead a team of 20 volunteers to structure, publicize, and organize entrepreneurial events on campus and actively contributed to building a startup ecosystem by connecting students with mentors, investors, and other valuable resources

Technical Skills

- **Languages:** C, C++, CUDA, Rust, Python, SQL, JavaScript, Bash, Verilog HDL, Assembly Language, MIPS
- **Development:** Typescript, React.js, Next.js, Node.js, Express.js, MongoDB, React Native, HTML, CSS
- **Libraries:** Sklearn, NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, Keras, Seaborn, OpenAI
- **Tools:** AWS, LaTeX, Docker, SQLite, PostgreSQL, NVBit, Git, Tailwind CSS, Bootstrap, Ghidra

Key Courses Undertaken

* Ongoing ** Online

Logic for CS	Linear Algebra and ODE	Introduction to Electronics
Fundamentals of Computing	Data Structures and Algorithms	Discrete Mathematics for CS
Deep Learning Specialization**	Programming with Java Script**	Multivariate Calculus
Computer Organization	Software Development Operations	Probability for CS
Operating Systems	Theory of Computation	Introduction to ML
Linux Kernel Programming	Parallel Computing	Compiler Design
Computer Networks	Programming for Performance	Principles of DBMS
Intro to RL	Computer Architecture	Big Data Visual Analytics

Miscellaneous

Volunteer Dec'22 - Mar'23
Student Placement Office, IIT Kanpur
Assisted in the placement drive of over 1000 students, ensuring a effective and hassle free process

Inter-IIT Tech Meet 11.0 Feb'23
Participated in Inter College Technical Meet in Cyber Security Problem Statement

Takneek'23 Mar'23
Mentored a team of 10 students in Inter-Hall Competitions